

★ WEEK 16 · FINAL

Bringing It All Together

The final project — one honest, end-to-end pipeline

Expanded edition · 38 slides · with the project skeleton and a full pre-flight kit

No new theory this week. Instead, you'll combine everything into a single project and learn what separates a real, trustworthy result from a flashy but hollow one.

Where We Are Going Today

This is your victory lap — and your launch pad. Let's make it count

1

Look back at the whole journey

16 weeks, four phases. The handful of lessons that matter more than any single algorithm.

2

The end-to-end pipeline

The five stages every project needs — and why the model is only one of them.

3

Your final project

Pick a track, follow the checklist, and present something you can stand behind.

By the end you will have priced an accuracy trap, a luck bar, a cost drag — and scored a project like a grader.

1

PART 1 • THE JOURNEY

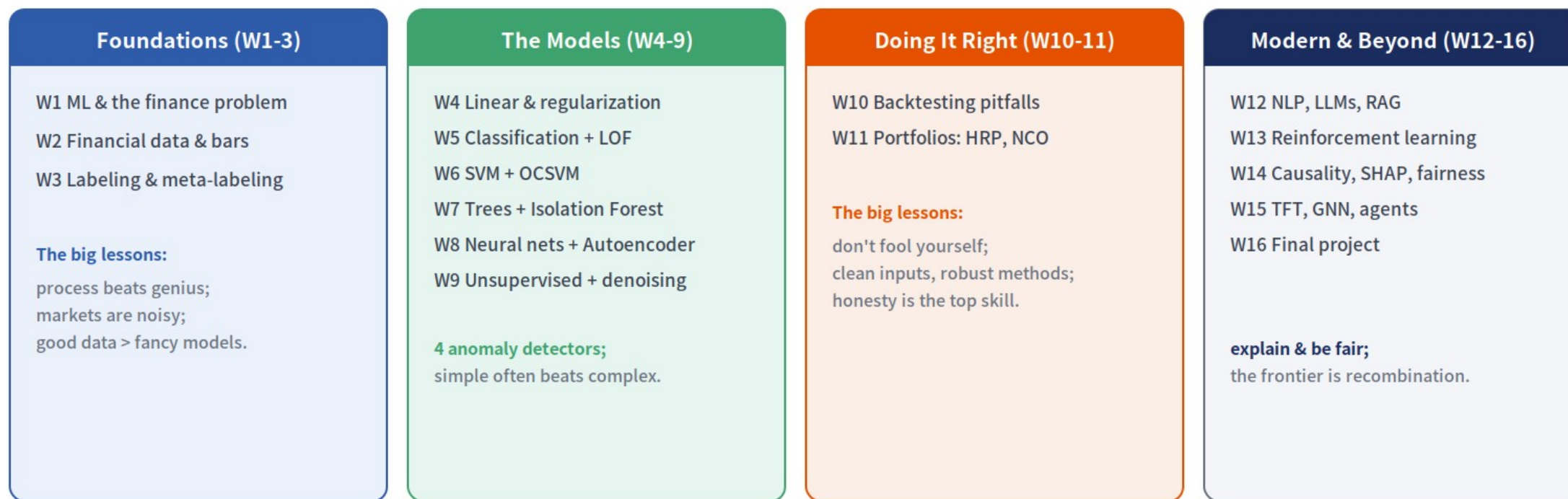
The Whole Journey

Sixteen weeks, a few big lessons

Everything You Learned, on One Page

Four phases: foundations, models, doing-it-right, modern and beyond. From 'what is ML?' to a complete, honest pipeline.

The Whole Journey — 16 Weeks at a Glance



From "what is ML?" to building a complete, honest, explainable pipeline. You made it all the way.

The Lessons Bigger Than Any Algorithm

Forget the formulas — remember these

1

Process beats genius (Week 1)

A disciplined, repeatable process matters more than one brilliant idea. The market humbles geniuses.

2

Good data beats fancy models (Weeks 2–3, 9)

Clean inputs, careful labels, and denoised correlations decide outcomes more than the model choice.

3

Honesty is the top skill (Weeks 10, 14)

Don't fool yourself: test without leakage, discount luck, explain decisions, and check fairness.

If you remember nothing else: be disciplined, respect the data, and stay honest with yourself.

Your Toolkit, by Job

Which tool for which problem

Predict a number or a class

- Linear / Ridge / Lasso (W4) — simple, explainable.
- Logistic, KNN, SVM (W5–6) — classify.
- Trees, Random Forest, XGBoost (W7) — tabular king.
- Neural nets (W8) — complex patterns, lots of data.

Find structure or act

- K-Means, PCA, denoising (W9) — group & simplify.
- Four anomaly detectors (W5–8) — spot the odd.
- HRP / NCO (W11) — robust portfolios.
- RAG (W12), RL (W13), agents (W15) — read & act.

Start with the simplest tool that could work; reach for complexity only when it earns its keep.

The Three Apps That Ran Through It All

Every method served one of these

Fraud detection

score & explain

Classifiers (W5–8) score each transaction; four anomaly detectors flag the unusual; SHAP (W14) explains every alert; a cascade screens fast, then scores sharply.

precision under imbalance

Finance chatbot

read & ground

Neural nets (W8) became LLMs (W12); embeddings + RAG ground the answers; agents (W15) let it act, not just talk. Always cite the source.

no hallucinated numbers

Trading strategy

signal & discipline

Signals (W4–8), clean covariance (W9), robust portfolios via HRP/NCO (W11), purged CV + Deflated Sharpe (W10), smart execution with RL (W13).

honest before clever

One toolkit, three products — the same ideas, pointed at different problems.

Sixteen Weeks, in Numbers

The shape of what you built

16

Weeks — foundations to frontier, no skipped steps

3

Apps that every single method had to serve

5

Pipeline stages — and the model is only one of them

48

Hand-calc problems: every number checked by pencil first

The pencil was the point: you never trusted a machine you couldn't check.

2

PART 2 • THE PIPELINE

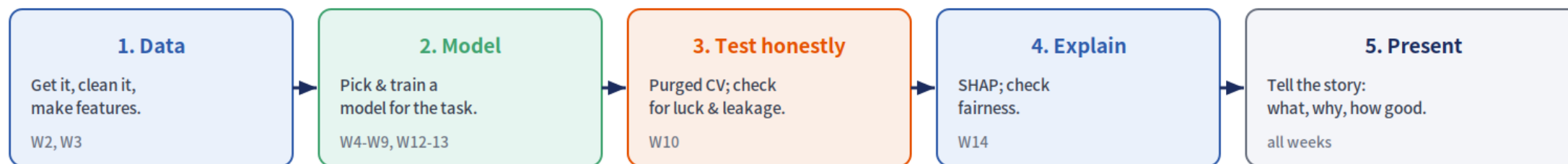
The End-to-End Pipeline

Five stages every project needs

A Project Is Five Stages, Not One Model

Data, model, honest testing, explanation, presentation. The honest testing and clear explanation are what earn trust.

Your Final Project — One End-to-End Pipeline



A complete project uses EVERY phase — not just a clever model

The model is just one box of five. Graders (and real employers) care most about honest testing and clear explanation.

A simple model, tested honestly and explained well, beats a fancy model you can't trust or justify.

Pick ONE of the three apps: fraud detection, a finance chatbot, or a trading strategy.

The Five Stages, as a Submission Checklist

Each stage has a deliverable — and a week that governs it

Stage	The deliverable	Governed by
1 · Data	source named, cleaning shown, labels justified	W2–W3
2 · Model	a simple baseline FIRST, then any upgrade	W4–W9
3 · Honest test	purged/time-aware CV, luck check, right metric	W10, W5
4 · Explain & fair	importance/SHAP + a per-group audit	W14
5 · Present	the story, the score, and the LIMITS — in that voice	W16

Print this. Every deliverable missing is a stage you'll be asked about in the Q&A.

The Five Most Common Project Mistakes

Avoid these and you're ahead of most

What sinks projects

- Jumping to a fancy model before cleaning the data.
- Leakage: a result too good to be true (it is).
- Reporting accuracy on imbalanced data (Week 5).
- No explanation; ignoring trading costs or fairness.

What graders reward

- A clear question and honest evaluation.
- Purged CV and a sanity check for luck.
- SHAP or importance; a fairness check across groups.
- An honest discussion of limitations.

Most points are lost on testing and explanation — not on picking the 'wrong' model.

The Accuracy Trap, Priced

1,000 transactions, 20 frauds. Your model: TP 8, FP 2, FN 12, TN 978 (full version: problem set)

Metric	Arithmetic	Value	Reading
Dumb baseline accuracy	$(980 + 0) / 1000$	98.0%	predict 'no fraud' always
Your model's accuracy	$(8 + 978) / 1000$	98.6%	barely beats doing nothing
Precision · Recall	$8/10 \cdot 8/20$	$0.80 \cdot 0.40$	the real picture appears
F1	$2 \cdot (0.8 \cdot 0.4) / (0.8 + 0.4)$	0.533	the honest headline number

On 2% fraud, accuracy grades the class imbalance, not the model. Week 5's oldest trap — and still the most submitted.

The Luck Bar, by Hand

You tried 20 strategy variants; the best backtest Sharpe is 1.2. Each junk Sharpe estimate has std 0.5 (full version: problem set)

Question	Arithmetic	Answer
Expected best Sharpe of 20 JUNK strategies	$0.5 \times \sqrt{2 \ln 20} = 0.5 \times 2.448$	≈ 1.22
Your best backtest	reported	1.20
Verdict	1.20 vs the 1.22 luck bar	indistinguishable from luck

Searching 20 variants manufactures a Sharpe ≈ 1.2 from pure noise. Report the number of trials with every result — Week 10's law.

Costs, Before the Applause

Trading track: edge 30bp per trade, cost 12bp, 40 trades a year, volatility 10% (full version: problem set)

Line	Arithmetic	Value
Gross return	$30\text{bp} \times 40$	12.0%
Net return	$(30 - 12)\text{bp} \times 40$	7.2%
Sharpe, gross $\rightarrow$ net	$12/10 \rightarrow 7.2/10$	$1.20 \rightarrow 0.72$
Breakeven cost	edge itself	30bp — above it, the strategy IS the broker's

Two-fifths of the story was costs. A trading project that reports gross numbers has skipped stage 5's honesty.

How a Great Project Is Judged

Trustworthiness counts far more than raw accuracy. Anyone can chase accuracy; you were taught to build something trustworthy.

What Makes a Great Project (and How It's Judged)

It's **NOT** about the highest accuracy. It's about doing the whole thing well:



Anyone can chase accuracy. The course taught you to build something trustworthy — that's what counts.

Score Three Projects Like a Grader

Weights: data .20, model .15, test .30, explain .20, present .15 — test + explain carry half (full version: problem set)

Project (scores 0–10 per stage)	Weighted arithmetic	Total
A · fancy model (6, 10, 3, 2, 8)	$1.2 + 1.5 + 0.9 + 0.4 + 1.2$	5.20
B · simple, honest (8, 6, 9, 9, 7)	$1.6 + 0.9 + 2.7 + 1.8 + 1.05$	8.05
C · middle road (7, 8, 6, 5, 6)	$1.4 + 1.2 + 1.8 + 1.0 + 0.9$	6.30

B beats A by 2.85 points with a plainer model — the 50% sitting on test + explain decided it. That is the whole rubric.

The Four Anomaly Detectors You Met

A running thread from Week 5 to Week 8

1

LOF (Week 5) and OCSVM (Week 6)

LOF spots a loner by low local density; OCSVM draws a fence around 'normal'. Distance- and boundary-based.

2

Isolation Forest (Week 7)

Anomalies take very few random cuts to isolate. Fast and scalable — great for large data.

3

Autoencoder (Week 8)

Trained on normal data only; weird points rebuild badly (big error). The deep-learning detector.

No single detector wins — real systems run several together, each covering the others' blind spots.

3

PART 3 · YOUR PROJECT

Your Final Project

Pick a track and build it

Three Tracks — Pick the One You Love

Fraud detection, a finance chatbot, or a trading strategy. Choose the problem that excites you — you'll do your best work on it.

Pick Your Project — Three Tracks to Choose From

A · Fraud Detection

Goal: flag suspicious transactions.

Use:

classifiers (W5-8), the four anomaly detectors, SHAP (W14).

Watch out for:

rare-event imbalance, leakage, fairness across groups.

Good for: classification & explainability.

B · Finance Chatbot

Goal: answer questions from docs.

Use:

embeddings + RAG (W12), maybe an agent loop (W15).

Watch out for:

hallucination — always ground answers and cite the source.

Good for: NLP & retrieval.

C · Trading Strategy

Goal: a tested, robust strategy.

Use:

signals (W4-8), portfolios (W11), purged CV + Deflated Sharpe (W10).

Watch out for:

overfitting, luck from many tries, unrealistic trading costs.

Good for: honest backtesting.

A Concrete Starting Point per Track

Data, baseline, honest test, explanation — filled in, so day 1 is building, not deciding

	Fraud	Chatbot	Trading
Data	\$ 5 set (2% positives)	a small filing/FAQ corpus you build	\$ 2/ \$ 3b price & panel loaders
Baseline	logistic regression	retrieval-only (no generation)	1-factor linear signal
Honest test	stratified CV + F1 / PR-AUC	held-out Qs; % answered WITH source	purged CV + Deflated Sharpe
Explain / fair	SHAP + per-group audit	citation rate; NOT-FOUND rate	top signals + net-of-cost result

Beat your baseline honestly or explain why you couldn't — both are A-grade sentences.

What You Must Deliver

A short, clear checklist for every track

1

A clear question and clean data

State exactly what you're predicting or building, and show how you prepared the data (Weeks 2–3).

2

A model, tested honestly

Train a model (any from the course), then evaluate with purged CV and a check for luck (Week 10).

3

Explanation, fairness, and an honest write-up

Use SHAP or importance, check fairness across groups (Week 14), and discuss limitations frankly.

A simple model done right is an A. A fancy model you can't trust or explain is not.

A Suggested Two-Week Plan

Pace yourself — don't leave testing to the end

Week one — build

- Day 1–2: pick a track, get and clean the data.
- Day 3–4: build features; train a first simple model.
- Day 5: get an honest baseline score (purged CV).
- Don't polish yet — get end to end working first.

Week two — harden & explain

- Improve the model; re-test honestly each time.
- Add SHAP/importance and a fairness check.
- Write up results AND limitations.
- Rehearse the story you'll present.

The biggest mistake: two weeks on the model, ten minutes on the testing. Flip that.

How to Present It Well

A clear story beats a wall of numbers

Tell a story in this order

- 1. The question: what are you predicting or building?
- 2. The data: where it came from, how you cleaned it.
- 3. The honest result: your score, and how you tested it.
- 4. The why: which clues drove it (SHAP/importance).
- 5. The limits: what you'd fix with more time.

Earn trust

- Say 'here's what didn't work' — it builds credibility.
- Name your number of trials next to any backtest.
- Show the fairness table without being asked.
- Answer 'why?' with the same calm as 'how much?'.

The presenter who volunteers the caveats owns the room; the one who hides them loses it in Q&A.

A Ten-Minute Presentation, Minute by Minute

The story slots, with the clock attached

Minutes	Slot	The one job of this slot
0–1	the question	one sentence; the audience must be able to repeat it
1–3	data & method	source, cleaning, the model in one diagram
3–6	the honest result	metric choice defended; CV scheme named; trials counted
6–8	the why & the fair	top clues + the per-group table
8–10	limits & next	what you'd fix; one concrete next step; stop on time

Six of ten minutes on honesty and explanation — the same 50% the rubric pays for.

The Four Gates Before You Submit

Answer all four in writing — your future Q&A is hiding in them

1 · Leakage gate

W10

Could tomorrow's information touch today's features or CV split? Name your purge.

'too good' is a symptom

2 · Luck gate

W10

How many variants did you try? Compare your best against the $\sqrt{(2 \ln N)}$ bar before celebrating.

count trials in writing

3 · Explain gate

W14

Can you show which clues drove the headline result — per prediction, not just on average?

SHAP or importance, shown

4 · Fairness gate

W14

Did you audit the metric BY GROUP? A blended number can hide a 25pp gap in plain sight.

the per-group table

A project that passes all four gates has nothing to fear from any examiner — including the market.

The Questions Everyone Asks

Straight answers, in the course's own voice

Question	Answer
Can I use Claude / AI to help?	Yes — Lab 3 rules: it explains and extends; you run, read, and defend every line.
Is a simple model enough?	A logistic regression tested honestly outscores an unexplained XGBoost. See the rubric arithmetic.
My result is negative — do I fail?	No. 'The signal did not survive costs and the luck bar' is an A-grade sentence, with evidence.
Can I change track midway?	Yes, before week two. A working simple project beats a broken ambitious one.
What sinks projects fastest?	Accuracy on imbalanced data, uncounted trials, and gross-of-cost returns — the three minis of today.

When in doubt, choose the honest sentence over the impressive one — that is the entire course in one rule.

Grader's Instinct — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

The submission says...	Your grader's note
'99% accuracy on fraud detection'	2% positives — demand F1/PR-AUC; the dumb baseline scores 98%
'Sharpe 1.2 after extensive experimentation'	how many trials? 20 junk variants also produce ≈ 1.22
'12% return' (trading track)	gross or net? At $12\text{bp} \times 40$ trades, 4.8% of it is costs
'we used XGBoost' (and nothing else)	where's the baseline, the explanation, the per-group table?
'the model slightly underperformed; here's why'	honest, evidenced, explained — this is what an A reads like

If you graded all five correctly, you are ready to grade your own project — do it before we do.

4

PART 4 · HANDS ON

Hands-On

A starter skeleton and where to go next

Getting Started

Use any data section from the companion notebook

Reuse what you know

- Trading track: the price/panel loaders (§ 2, § 3b).
- Fraud track: the classification set (§ 5).
- Chatbot track: a small text corpus you build inline.

Start simple, then grow

- Get the full pipeline running with a basic model first.
- Only then try a fancier one — and re-test honestly.
- Commit often; keep a working version at all times.

[CODE] colab_project.py · the five-stage skeleton

```
# 1 DATA    -> load, clean, make features (Weeks 2-3)
# 2 MODEL    -> train a simple model first (Weeks 4-9)
# 3 TEST     -> purged CV; check for luck/leakage (Week 10)
# 4 EXPLAIN  -> importance/SHAP; fairness check (Week 14)
# 5 PRESENT  -> results, limitations, and the story
from sklearn.datasets import make_classification as mc # guide §5: X = clues, yb = label
X, yb = mc(20000, 20, n_informative=6, weights=[.99, .01], random_state=42)
```

Hands-On — A Minimal Honest Pipeline

All five stages in one tiny example (fraud track)

[CODE] colab_project.py · minimal honest pipeline

```
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.metrics import f1_score

# 2 MODEL + 3 honest TEST (cross-validated F1, not accuracy)
cv = StratifiedKFold(5, shuffle=False) # keep it honest
m = RandomForestClassifier(n_estimators=200)
scores = cross_val_score(m, X, yb, cv=cv, scoring='f1')
print('honest F1:', scores.mean().round(3))

# 4 EXPLAIN
m.fit(X, yb)
print('top clues:', np.argsort(m.feature_importances_)[-5:])
```

Your task (step by step)

- Load X, yb (setup guide § 5).
- Train a simple model.
- Score it with TIME-AWARE / honest CV.
- Print the top clues (explanation).
- Then: swap in your own track and grow it.

Claude as Your Project Copilot

AI is allowed on the final project — under the same house rules as all semester

[INPUT] Prompt · Claude (paste with your project draft)

You are my final-project reviewer for a financial ML course. My draft code and write-up are above.

- 1) Verify my three hand-calcs: (a) TP 8, FP 2, FN 12, TN 978 -> F1 = 0.533 while accuracy 98.6% vs the 98.0% dumb baseline; (b) luck bar for 20 trials, sd 0.5 -> 1.22; (c) 30bp edge, 12bp cost, 40 trades -> net 7.2%, Sharpe 1.2 -> 0.72.
- 2) Audit my project against the four gates: leakage, luck (count my trials), explanation, per-group fairness - cite the exact line of my code where each passes or fails.
- 3) Score me on the rubric (.20/.15/.30/.20/.15), one sentence of evidence per stage.
- 4) Rewrite my weakest paragraph in the honest voice: claims sized to evidence, trials counted, limits named.
- 5) List the 3 questions a tough examiner will ask, with the answers my current draft supports.

House rules for AI use

- Allowed: explain, debug, extend, review your work.
- Required: run it, read it, defend every line.
- Forbidden: submitting anything you cannot explain.
- The viva voce is yours alone — prepare for it.

Where to Go After This Course

You have the map; here's how to keep climbing

Keep learning

- Read the source texts (Hull; López de Prado).
- Reproduce a paper end to end.
- Enter a Kaggle competition — but test honestly.
- Go deeper on the topic that gripped you most.

Build a portfolio (of work!)

- Turn your project into a clean public write-up.
- Show the PROCESS, not just the result.
- Employers value honest evaluation and clear thinking.
- Your discipline is your edge — show it off.

The tools will keep changing; the principles you learned here will not.

One Last Thing

The mindset that outlasts every algorithm

2.3

You now have both the tools AND the judgment to use them well. That's what this course was for.

Final Comprehension Check

Two questions that cover the whole finale

Q1 Fraud data, 2% positives. Your model reports 98.6% accuracy. You should:

- A** Celebrate — that's an A
- B** Compare to the 98% dumb baseline and report F1 (0.533) ✓
- C** Add more features
- D** Switch to a neural net

Answer: The all-negative baseline already scores 98%. On imbalanced data the honest headline is F1 or PR-AUC — Week 5's law.

Q2 Which project scores highest on the rubric?

- A** Fancy model, thin testing (5.20)
- B** Simple model, honest test + full explanation (8.05) ✓
- C** They tie
- D** Whichever has the best accuracy

Answer: Test + explain carry 50% of the weight. B's 2.7 + 1.8 on those two stages alone nearly matches A's entire total.

The Final Week on One Page

Photograph this slide — it is the project brief

Item	One line to remember
The pipeline	data → model → honest test → explain & fair → present; the model is ONE box.
The accuracy trap	98.6% vs a 98.0% dumb baseline; $F1 = 0.533$ is the honest headline.
The luck bar	best of 20 junk trials $\approx 0.5 \cdot \sqrt{2 \ln 20} = 1.22$ — count your trials in writing.
Cost honesty	$(30 - 12)\text{bp} \times 40 = 7.2\%$ net; Sharpe $1.2 \rightarrow 0.72$. Report net, always.
The rubric	weights .20/.15/.30/.20/.15 — honest testing + explanation carry half.
The four gates	leakage, luck, explanation, per-group fairness — pass all four, then submit.

A simple model done right is an A. Build something you can defend with a straight face.

The Course in Three Sentences

Sixteen weeks, one complete picture: how to do financial machine learning the right way

1

You learned the tools

Models from linear to deep, anomaly detectors, portfolios, NLP, RL, and the frontier — all from the ground up.

2

You learned to use them honestly

Clean data, purged testing, deflated luck, clear explanations, and fairness — the discipline of real practice.

3

You learned what matters most

Process over genius, data over models, honesty over a pretty backtest. That judgment is your real takeaway.

Keep the habit that built all of it: pencil first, machine second, honesty always.

Your Assignment, in One Slide

Good luck — build something you're proud of, and that you can defend with a straight face

1

Pick one track

Fraud detection, a finance chatbot, or a trading strategy — whichever excites you most.

2

Build the full five-stage pipeline

Data, model, honest testing, explanation and fairness, then a clear presentation. The model is just one box.

3

Stand behind your result

Report it honestly, discuss its limits, and be ready to explain every decision. Trustworthy beats flashy.

Thank you for sixteen weeks of hard work. Now go build.